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This implements Air Force Instruction (AFI) 21-101, *Aircraft and Equipment Maintenance Management.* It provides guidance and procedures on management and reporting for requesting, funding, executing, and reporting Test Program Software (TPS) maintenance engineering projects. It applies to civilian and military members assigned to Air Force Sustainment Center (AFSC), 448 Supply Chain Management Wing (SCMW) and Supply Chain Management Group (SCMG) personnel consisting of Engineers, Equipment Specialists (ES), Program Managers (PMs), Supply Planners, Contracting Officers, Financial Specialists, Software Maintenance Organization points of contact personnel who participate in TPS maintenance requirement process. This publication may be supplemented at any level, but all supplements must be routed to the Office of Primary Responsibility (OPR) listed above for coordination prior to certification and approval. Refer recommended changes and questions about this publication to the Office of Primary Responsibility listed above using the Air Force Form 847, *Recommendation for Change of Publication*; route AF Form 847 from the field through the appropriate chain of command. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with AFI 33-322, *Records Management and Information Governance Program*, and disposed of in accordance with the Air Force Records Disposition Schedule (RDS) located in the Air Force Records Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. Major changes include: adding specific inclusive verbiage on Maintenance Planning & Execution (MP&E), changed the former General Information Chapter to Overview, and added Software Maintenance as its own chapter.

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1. Overview.

1.1. TPS requirements. Provide for software maintenance engineering, which include; but is not limited to investigations, analyses, design, code, and test. TPS requirements also include development of testing capability, development of Interface Test Adapter (ITA), and development/update of drawings and manuals as a direct result of incorporated changes that enable continued TPS capability. Typically, TPS requirement initiatives are to resolve documented Deficiency Reports (DR) that come from a variety of sources (e.g., the operator, maintainer, mishap investigation reports, and/or developer). TPS requirements may also be used for restoration of TPS capability driven by a Consolidated Sustainment Activity Group (CSAG) project for a redesign of a CSAG item (more appropriate as a phase of a Consolidated Sustainment Activity Group – Supply (CSAG)-S project). If requirements are left unresolved, these deficiencies could result in deficient assets being returned into supply resulting in potential mission incapability and/or Safety of Flight concerns to Air Force (AF) weapon systems/personnel.

1.2. Background. The processes described in this instruction are the results of a Tinker Integrated Product Team (IPT) consisting of focal points from the 448 SCMW and Oklahoma City-Air Logistics Center (OC-ALC), formerly known as 76 Maintenance Wing (MXW). The IPT was formed to develop an operational instruction for all sites to clarify CSAG-S software maintenance engineering requirements and funds execution processes for UUT/TPS program in response to AF Audit Agency Audit, F2010-0030- FCT000, Project # 2009-FC2000-0019.004. Due to organizational changes, responsibility of the UUT program (contract and organic) was delegated to Strategic Alternate Sourcing Program Office (SASPO) for management and maintenance of the requirements process. On 30 November 2011, an extension was granted to postpone publication until 01 January 2015, allowing the re-establishment of Job Order Master Production System (G004L) as the funds tracking mechanism process, slated for FY14 implementation. Continuous process improvements in conjunction with G004L implementation have been incorporated in this instruction.

2. Roles and Responsibilities.

2.1. 448 SCMW/EN (Technical Director) shall:

2.1.1. Approve consolidated Wing Project Requirements List.

2.1.2. Provide guidance to DoEs and Strategic Sourcing Program regarding technical issues and policy.

2.2. Supply Chain Management Group/Engineering (SCMG/EN) Flight Chiefs shall:

2.2.1. Review, validate and approve all TPS software maintenance engineering projects for accuracy, validity and scope prior to submission for signature.

2.2.2. Submit completed projects to Strategic Sourcing Program Office for validation and change to 2.4 status, once the DoE review panel has approved the project.

2.2.3. Ensure Requirement Generators (RG) respond in a timely manner during Strategic Sourcing Program Office validation.

2.2.4. Ensure RGs update TPS expense reporting at least monthly in Maintenance Engineering Project Sustainment Tool (MPST), and file maintain workload requirements in MP&E/R2D2 to support Workload Review.

2.2.5. Ensure all appropriate personnel necessary for monthly DoE project status meetings to include PMs, Software Organization (SWO), Financial Management (FM), Contracting (PK), etc., have been notified.

2.2.6. Ensure status updates for monthly DoE project status meetings reflect latest project development/execution in MPST prior to meeting.

2.3. Supply Chain Management Groups (SCMG) DoEs shall:

2.3.1. Establish and chair a review panel to formally evaluate, validate, and approve projects at site level.

2.3.2. Disapprove projects that are not within scope, incomplete, or lack adequate justification and/or documentation. **Note**: In the event DoE is not available due to extended absence or special assignment, the DoE may temporarily appoint an alternate. Every project approved in the appropriate application/database will also attain full coordination within MPST (see **Table 5**) before submitting to SASPO for final project validation.

2.3.3. Collect, review, finalize, sign, and provide a list of their respective Group's projects and forward to Strategic Sourcing Program Office no later than (NLT) end of month (EOM) February (organic) and NLT 15 March (contract), or both may be submitted in February of the current year, in response to SASPO data call.

2.3.4. Determine and coordinate with Strategic Sourcing Program Office and 448 SCMW/FM on distributing funds proportionately when full funding is not received.

2.3.5. Require project RGs to form an Integrated Product Team (IPT) to ensure requirement's planning, continuity, and development.

2.3.6. Chair monthly DoE project status meetings facilitated by Strategic Sourcing Program ensuring timely achievement of major milestones. At the discretion of the DoE, monthly meetings may be suspended until late in FY (4th Quarter).

2.4. Supply Chain Management Squadron CC/CL/DD shall:

2.4.1. Review TPS software maintenance engineering projects as applicable.

2.4.2. Coordinate or non-coordinate projects in MPST.

2.5. CSAG – Maintenance Division (CSAG-MD)/Software Engineering Group (SWEG)/Software Maintenance Source shall:

2.5.1. Submit Software Deficiency Report (SWDR) into Joint Deficiency Reporting System (JDRS) or other alternate means of tracking as deficiencies are identified for Requirements Generator (RG) review.

2.5.2. Perform TPS deficiency investigations and analysis to identify resolutions as required.

2.5.3. Predict level of effort required and provide basis of cost estimate for TPS maintenance support, i.e., negotiated spreadsheet, ROM (Rough Order Magnitude), for the projected year by project.

2.5.4. Perform TPS software maintenance corrective actions on approved SWDR.

2.5.5. Perform Software Control Center (SCC) functions as necessary, which include storing, duplicating, and distributing TPS software as requested.

2.5.6. Provide monthly status reports for each Program Control Number (ID/PCN) to SCMG RGs.

2.5.7. Support monthly Director of Engineering (DoE) project status meeting and provide required support as needed.

2.5.8. Report shortfalls, excess funding (hours and dollars) to RGs. Reference **Paragraph 8.3** for processing instructions.

2.5.9. Support internal and external TPS audits as required.

2.5.10. Report estimated dollars and hours needed to complete work on open Job Order Number (JONs) prior to end of year closeout. A schedule completion date will be required on all open JONs.

2.5.11. Monitor all open JONs AFMC206, *Temporary Work Request*, Production #, T-jobs, etc., until work- load is complete.

2.5.12. Execute close-out actions for all completed JONs.

2.5.13. Provide close-out status to RGs. Reference MPST User's Guide for further step-by-step guidance.

2.6. Requirements Generator (RG). Typically SCMG staff members that include, but are not limited to: engineers, program managers, equipment specialists, and supply planners. In MPST, the RG may also be known as the originator. RGs shall:

2.6.1. Determine the appropriate software maintenance group, understand the level of effort required, establish requirements to be accomplished, attain a schedule of work performance and ensure requirements are submitted on time.

2.6.2. Submit TPS software maintenance engineering project for upcoming FY in MPST to maintain a sustained testing capability for UUTs.

2.6.3. Establish an organic, contract, or DMISA (interservice/intraservice) TPS PCN and ensure requirements are file maintained within Maintenance Planning and Execution (MP&E), and meet Requirements Review and Depot Determination (R2D2) forecasts. Reference AFMCMAN 20-102, *Maintenance Planning and Execution System (MP&E) (D363).*

2.6.4. Collaborate with R2D2 Repair Program Manager (RPM) on hours file maintained for TPS projects.

2.6.5. Attend and provide project status updates at monthly DoE project status meetings for contract TPS and quarterly project status meeting for organic TPS.

2.6.6. Accomplish funding action required to commit/award workload.

2.6.7. Track requirements through the project identification number assigned by MPST and corresponding PCN assigned by MP&E.

2.6.8. Consult with a team of planners, schedulers, software developers, logisticians, engineers, contracting, financial specialists, and other technical personnel as required for planning purposes.

2.6.9. Ensure all required project elements have been provided and uploaded in MPST prior to submittal to Strategic Sourcing Program. Reference **Table 3**.

2.6.10. Monitor and validate periodic reports from designated software maintenance source to verify that only approved TPS updates are being worked and reasonable hours are charged to the correct MPST project ID/PCN.

2.6.11. Approve SWDR completion by the acceptance of the corrective action.

2.6.12. Ensure TPS software maintenance engineering requirements within MPST, MP&E, and other applicable systems reflect the most current data.

2.6.13. Initiate AFMC206 as early as 1 October for each TPS project.

2.6.14. Report shortfalls, excess funding (hours and dollars) to Strategic Sourcing Program and document in MPST message board. Reference **Paragraph 8.3** for processing instructions.

2.6.15. Support internal and external TPS audits as required.

2.6.16. Request adequate funds, hours and schedule for open JONs are available prior to end of year closeout.

2.6.17. Ensure projects requiring additional information during Strategic Sourcing Program validation are returned promptly.

2.6.18. Maintain supporting documentation for each project until one year after completion of requirement. Reference MPST User's Guide for further step-by-step guidance.

2.6.19. Obtain close-out report/status from Maintenance Source POCs for all projects and coordinate closure with Squadron, System Program Office/System Directorate.

2.6.20. Obtain coordination from MAJCOM on TPS projects in MPST

2.7. 429 SCMS/GUMD Strategic Alternate Sourcing Program Office (SASPO) shall: Provide enterprise wide guidance and program management support for sustaining engineering projects that improve safety, sustainability and reliability of Air Force weapon systems.

2.7.1. Provide Wing guidance and direction to SCMGs for development and execution of TPS software maintenance requirements.

2.7.2. Issue upcoming year requirement data calls (First of January) to 638/748/848 SCMGs and Air Logistics Complexes (ALCs) with a due date (EOM February for organic and EOM April for contracts) for receiving each center's validated TPS software maintenance engineering requirements.

2.7.3. Conduct review of projects to ensure requirements are accurately defined, rigorously supported, and properly documented.

2.7.4. Validate all consolidated Group TPS software maintenance engineering project requirements lists and provide an approved consolidated Wing project requirements list to 448 SCMW/FM NLT EOM May.

2.7.5. Request distribution of available funding for approved projects from 448 SCMW/FM.

2.7.6. Establish a TPS software maintenance engineering project comprehensive and dynamic list to be updated throughout the year with funded projects and those waiting to be funded and post to the 429 SCMS/GUMD SharePoint®.

2.7.7. Monitor execution to ensure timely commitments and obligations.

2.7.8. Make recommendations for application and removal of funds from projects.

2.7.9. Perform project assessments as necessary to ensure proper accounting of TPS software maintenance engineering requirements and funds execution.

2.7.10. Monitor the entire process and record periodic reports received from the RGs throughout the funds execution period.

2.7.11. Prepare program status updates for senior leaders and 448 SCMW/FM as required.

2.7.12. Call, invite, prepare and facilitate monthly DoE project status meetings as determined by Strategic Sourcing Program with input from DoEs. At the discretion of the DoE, monthly meetings may be suspended until late in FY (4th Quarter).

2.7.13. Conduct internal audit for past and present projects to ensure effective project management.

2.7.14. Ensure radio button for DoE report is selected yes after project is funded.

2.8. 448 SCMW/FM shall:

2.8.1. Act as a liaison between the Strategic Sourcing Program and AFSC/FM.

2.8.2. Provide a consolidated Wing Project Requirements List to AFSC/FM for budget submission.

2.8.3. Request funds from AFSC/FM.

2.8.4. Identify deadline for budget submittal.

2.8.5. Inform Strategic Sourcing Program and 448 SCMW-OLs (Operating Locations) of FY funding distribution levels received from AFSC/FM.

2.8.6. Notify Strategic Sourcing Program of surplus/fall-out funds available for realignment.

2.8.7. Ensure all funds requested are utilized in year of funding.

2.8.8. Attend and provide financial status updates at monthly DoE project status meetings.

2.9. AFSC/PZ shall (Contract TPS only):

2.9.1. Review and provide concurrence on annual program funds execution schedule.

2.9.2. Attend and provide contract award status updates at monthly DoE project status meetings as appropriate.

2.10. 448 SCMW/FMA-OL shall:

2.10.1. Monitor and track funds execution in the Metrics Financial Tool (MFT), Commanders Resource Information System (CRIS), FM Suite and Automated Project Order System (J025A) to ensure timely and accurate commitments/de-commitments and obligations/de-obligations of project funds. Reference **Paragraph 8**.

2.10.2. Obtain Funds Classification Reference Number (FCRN) for the PCN on each organic TPS requirement to track funding and load Program Element Code (PEC) code.

2.10.3. Submit fund cite build requests to the Accounting Liaison Office (ALO) for Out of Cycle requirements and loads funding.

2.10.4. Submit Project Order (AFMC181) for funds certification and coordinate funds on Purchase Request (PR) and AFMC206.

2.10.5. Save and load screenshots of the AFMC Forms 206 and AFMC Forms 181 to FMSuite for reconciliation purposes and to assist with Tri-Annual Review/Open Document Listing (TAR/ODL).

2.10.6. Manage and realign funding at location between Center distribution levels and individual requirements.

2.10.7. Support and provide financial status updates for DoE project status meetings.

2.10.8. Monitor and ensure accuracy in execution status as it occurs in the MPST and applicable financial tools depicting commitment and obligation for funded active sustaining engineering projects.

2.10.9. Track and report monthly forecast and subsequent variance analysis for obligations and expenses.

2.10.10. Ensure all excess funds/unliquidated obligations (ULOs) at contract completion are de-obligated in financial systems and reflected in MPST to ensure funds FY reconciliation. Notify Strategic Sourcing Program of action completion.

2.11. AFSC/LGPS shall:

2.11.1. Provide policy and guidance for TPS software maintenance program.

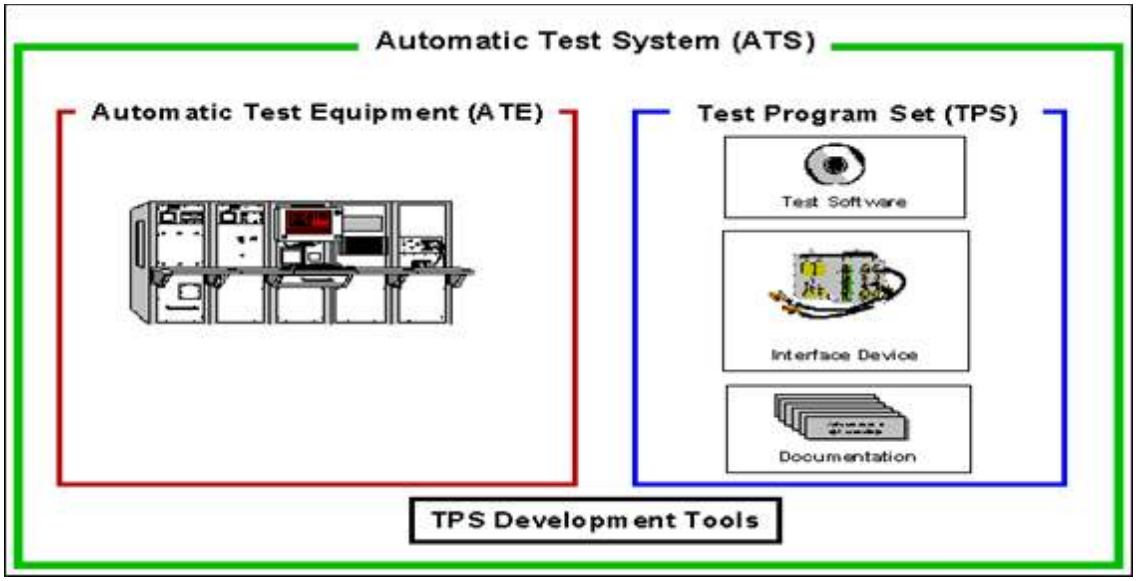
2.11.2. Provide timely resolution of TPS software maintenance concerns and program related issues.

3. TPS.

3.1. Software Maintenance. In accordance with (IAW) AFMAN 63-143, *Centralized Asset Management Procedures,* software maintenance is defined as the production effort required to design, code, test, and produce embedded weapon system and associated test system software. Software maintenance requirements are changes designed to correct errors or deficiencies, improve performance within existing specifications or adapt to a changing environment. These software maintenance requirements primarily fall into two categories: Depot Purchased Equipment Maintenance (DPEM) and CSAG UUT funded software. For this instruction, only software maintenance changes designed to correct errors or deficiencies for budget code 8 UUTs are applicable to CSAG –S UUT funded software. DPEM funded software is referred to as Program Office funded software, i.e., requirements that adapt to a changing environment,

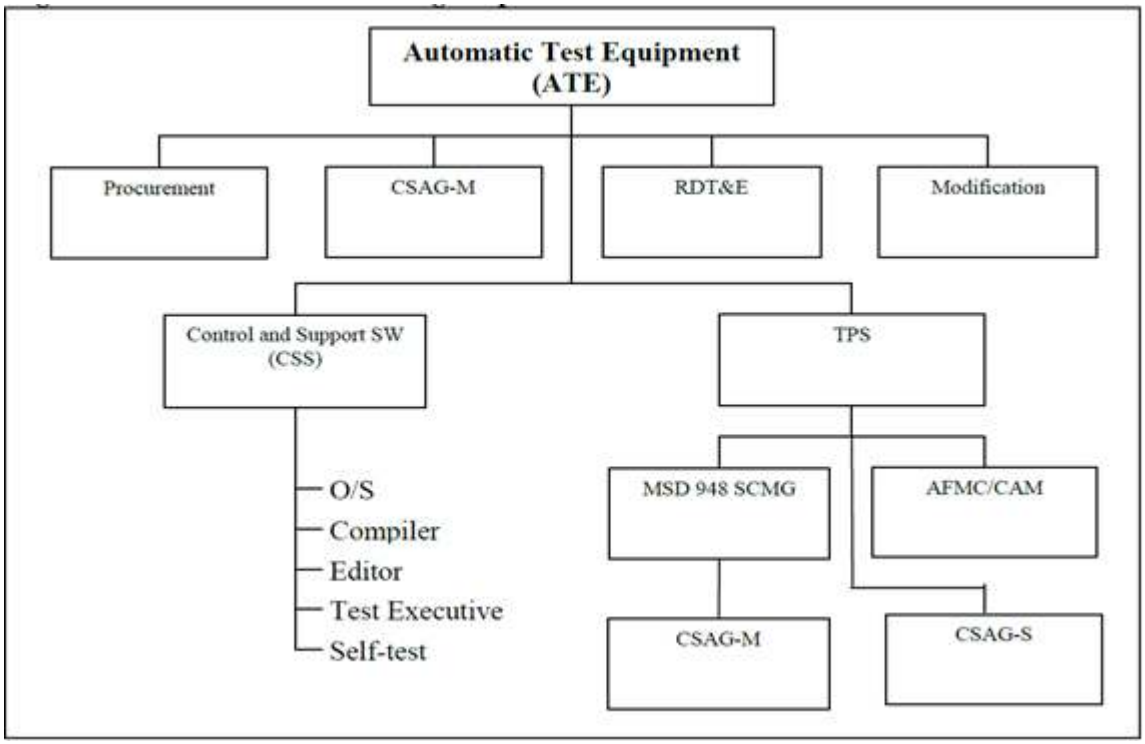
improve performance, etc. Software that the Program Office has direct responsibility for includes as a minimum, the following types of software: Operational Flight Programs (OFPs); TPS that test a system as installed on the aircraft at the organizational level; Software that supports a weapon system, and not used for the repair of items (i.e., Mission Planning, organizational level TPS). To determine which software requirements fall under CSAG-S UUT, one must first understand the elements of Automated Test System (ATS). ATS includes Automated Test Equipment (ATE) hardware, Control and Support Software (CSS), TPS, and UUT. **Figure 1** illustrates ATS system elements.

Figure 1. Elements of the Automatic Test System.



3.2. Types of ATS. Per AFMAN 63-143 Chapter 4.7, there are two types of ATS. One type of ATS is used to support CSAG-S managed UUTs; these are items that CSAG has responsibility for repair. The other type of ATS is used to test systems on the aircraft, these items are not CSAG-S managed, and are funded with Centralized Asset Management (CAM) dollars. Funding responsibility depends on which organization has responsibility for the item under test or repair. **Figure 2** Illustrates ATS software funding responsibilities. For further clarification on other than CSAG-S funding responsibilities, refer to AFMAN 63-143.

Figure 2. Illustrates ATS Software Funding Responsibilities.



3.3. Maintenance Engineering Funding Propriety. Precedence for funding propriety is stated within AFMAN 65-605."Use Defense Working Capital Fund (DWCF) to fund development, production, and maintenance engineering services that support an operational commodity item managed by the DWCF CSAG-S. Do not use the DWCF if the engineering effort results from a weapon system production or modification improvement."

3.4. CSAG-S TPS Funding Responsibility. CSAG-S funding requirements are captured within the depot line item repair costs and tracked through the CSAG accounting system. Software funded by CSAG-S UUT funding includes, as a minimum, TPS that require software maintenance due to existing baseline deficiencies for the repair of CSAG-S National Stock Number (NSN) items. A TPS consists of the software, ITA (i.e., hardware interface), and documentation used with ATE (usually in the form of a disk, cartridge, tape, etc.) to test and repair CSAG-S items. The software contains the test parameters to facilitate testing of items and subsystems. Software changes do not give the ATE new capability; they facilitate the application of the ATE in maintenance of the UUT. If there is a form, fit, function (FFF) or integration change to the test equipment hardware which may involve software changes to the TPS, the requirement is considered a modification and the applicable modification manager should be contacted regarding all software requirements for applicable modification funding, Research, Development, Test and Evaluation (RDT&E) or procurement funds. TPS re-hosting for the purpose of testing and repairing CSAG-S items and moving TPS to another existing ATE due to current ATE obsolescence issues should be funded by the account that funded the modification or procurement of the ATE. TPS projects fix existing software errors. Example: If a test parameter changes due to a modification of the UUT, TPS does not fund the change. If the parameter were entered incorrectly; then TPS does fund it. TPS project submittals supply formal documentation for software requirements at the NSN level by weapon system as

applicable. **NOTE: Tables 1 and 2** illustrate software maintenance considered to be funded or not funded by CSAG-S TPS.

3.4.1. ATE. If the test equipment requires a change (e.g., from Windows 98 to Windows XP operating system) (i.e., re-hosting), this change will be considered a DPEM software requirement, not funded by CSAG-S. If the TPS requires a change as a result of a test equipment change this will also be a DPEM software requirement, not funded by CSAG-S.

3.4.2. TPS. Software changes, containing test parameters for the testing of a CSAG-S item. The TPS includes the software, the ITA, and the technical data and/or documentation. The Project Manager, Engineer or Equipment Specialist in a Supply Chain Management (SCM) organization may initiate and submit a UUT project/requirement.

Table 1. Funded under CSAG-S TPS.

Software Maintenance includes, but is not limited to the following:
Deficiency investigations tied to CSAG-S NSN that has been identified as having a software deficiency, Incident Report (IR), Production Quality Deficiency Report (PQDR), DR, Software Trouble Report (STR), and SWDR.
Requirements analysis for resolution.
Software design within the TPS.
Coding of TPS software.
Module testing.
Integration testing.
Lab testing.
Aircraft Re-Test OK (RTOK) testing. RTOK applies to only LRU/SRUs tested OK on ATE but failed Re-Test on aircraft.
Interface Test Adaptors (ITA), which consist of wiring, cables, connectors, and passive and/or active components to provide a compatible connection between the UUT and the test equipment for the specific testing of CSAG-S items.
Configuration/Technical Data, which includes the cost of updating engineering data (e.g., production manual, documentation, source code, etc.), Technical Orders (T.O.s) and Time Compliance Technical Orders (TCTOs).
Cost of reproduction and distribution of the TPS software.

Table 2. Not Funded Under CSAG-S TPS.

Excluded from Software Maintenance, but is not limited to the following:
The cost of printing and distribution of manuals (includes Technical Orders, User manuals, etc.) is funded by Element of Expense Investment Code (EEIC) 594PO, T.O. funding.
Any travel required by CSAG - Maintenance Division (MD) personnel should be submitted through the SCM; and if approved, funded by CSAG - Supply Division (SD) 409 account. TPS projects do not fund temporary duty (TDYs).
TPS re-host to Capital Investment Program (CIP) funded CSAG-MD owned equipment or System Program Office (SPO) owned equipment.
ATE sustainment hardware and operational software. (To determine the appropriate funding source for ATS software support on Government Furnished Equipment (GFE), treat it as though the ATE were field equipment).
UUT component repair and ITA modifications not associated with TPS maintenance action.
Engine Test Cell TPS sustainment.

3.5. TPS Source Type and Direct Fund Cite. TPS software maintenance engineering requirements can be separated into three basic funding source categories, contract, organic and Depot Maintenance Interservice/Intraservice Support Agreement (DMISA).

3.5.1. CSAG-S funds are used to contract TPS software maintenance as a direct fund cite under Element of Expense Investment Code (EEIC) 56000. Contract work- loads are funded with Cost Authority (CA). Funds can be tracked in USAF Standard Base Level General Accounting and Finance System (H069/BQ) and Contract Asset Visibility (CAV) AF system. Requirements are completed via use of non-governmental organizations and resources. These requirements may continue to be worked and funds expended until the requirement has been completed.

3.5.2. CSAG-S funds are used to support organic TPS software maintenance as a direct fund cite under EEIC 54001. Organic projects funds can be tracked through the Automated Project Order (APO) System (J025A) form AFMC181, G004L, and Defense Industrial Financial Management System (DIFMS) systems. Requirements are completed via the use of governmental organizations and resources. For projects that are approved late in the year, these requirements can extend past year of obligation. The balance of funding required at the end of year (EOY) should equal the hours necessary to complete the open job order. Any new inductions beginning on or following 01 October of each FY should be funded with the fiscal year dollars assigned to that period.

3.6. DMISA Projects. DMISA projects are funded by CSAG-S, and EEIC 54002 will be used. DMISA is the formal agreement similar to a contract whereby one military Service (the Agent) agrees to provide depot maintenance support for another Service (the Principal). It may also be used when a Service is the Agent and another federal department agency or element thereof is the Principal. IAW OPNAVINST 4790.14B, *Joint Depot Maintenance Program*, DMISA shall be used for all multi-year inter-Service depot maintenance workload assignments. TPS projects where depot maintenance support is within the same service (Air Force) do not qualify as DMISA. IAW with AFI 65-118, *Air Force Purchases Using Military Interdepartmental Purchase Requests (MIPR)*, intra-agency requirements are obtained from an activity within Department of Defense (DoD). Example: An Air Force unit forwards a MIPR to the Army and an order is placed for supplies/services for the Air Force from an existing contract. The Army is the intra-agency assisting agency. Interagency requirements are obtained from an agency outside DoD. Example: Air Force unit forwards a MIPR to General Services Administration (GSA) and an order is placed for supplies/services for the Air Force from an existing contract. GSA is the interagency assisting agency. DD Form 448, *Military Interdepartmental Purchase Request,* will be used to process MIPRs.

3.7. Internal Audits. The Strategic Sourcing Program shall conduct an internal audit bi-annually of the CSAG-S/TPS software maintenance engineering projects, past and present, to ensure effective project management. A representative sampling of 5-7 projects per site, Oklahoma City (C), Ogden (OO), and Warner Robins (WR), for a total of 15 - 20 projects will be assessed. Audit assessments will be conducted by use of a local checklist and may vary according to focus areas of concern. Audits will be conducted annually as required. As of this instruction, bi-annual TPS audits shall commence one year after publication to allow familiarization and compliance of guidance. Reference **Attachment 2**, Sample Audit checklist.

4. Software Deficiencies.

4.1. UUT Deficiencies. A DR is generated by a user, and includes a detailed description of a problem encountered with a UUT. Once a deficiency report is established for a UUT testable item, a technical team (i.e., cognizant engineer, equipment specialist, and software engineer) reviews and investigates to determine root cause of the deficiency. Deficiency investigations tied to a CSAG-S Budget Code 8, National Stock Number (NSN), are a result of a software deficiency being identified through; IR, PQDR, DR, Operational Change Request (OCR), STR. Refer to T.O. 00-35D-54, *Technical Manual: USAF Deficiency Report, Investigation, and Resolution,* for guidance on reporting software deficiencies. **Note**: In this instruction, DR identified to be related to software will be called SWDR.

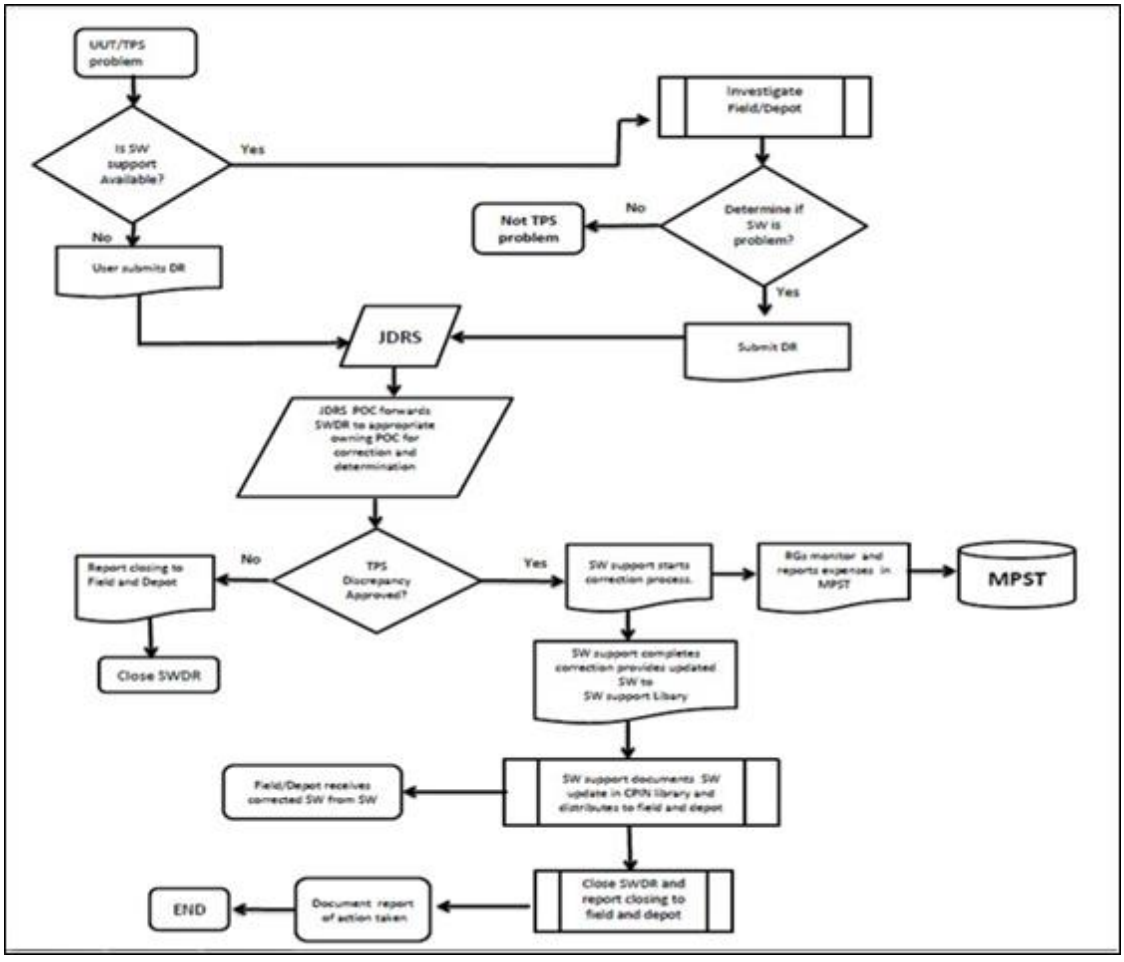
4.2. Isolation of Deficiency to AF TPS. Identified deficiency must be isolated directly to AF TPS. **Table 1** and **Table 2** illustrate what is authorized and not authorized to be funded by CSAG-S TPS

4.3. Establishing an SWDR. SWDR are typically identified either by Depot shops or Field units. Depot shops will usually request engineering support investigation prior to establishing an SWDR to confirm if the deficiency is software related. After investigation by a technical review team, if a deficiency is determined to be software related, depot shops will input SWDR in JDRS or other alternate means of tracking. Field units will submit SWDR prior to determining software deficiency by inputting SWDR through JDRS.

4.4. Correction of Deficiencies. As deficiencies are identified, software support personnel will start the correction process. After correction of the deficiency is completed, software support personnel will document the correction, update the Computer Program Identification Number (CPIN) library, provide distribution to field and depot, and close the SWDR. See **Figure 3** TPS Software Deficiency Correction Process Flow Chart.

4.5. Software Configuration Management Consideration. Once the TPS has been baselined and accepted by the government, all changes to the TPS are tracked through the Configuration Control Board (CCB) process. Software is given a unique configuration identifier CPIN and is tracked via the Enhanced Technical Management System (ETIMS). Changes to the TPS software are usually the result of a DR correction and result in a revision/change/update to the CPIN number. CPIN revisions must be approved by CCB before released for use. TPS changes may or may not get reported to Weapon System Chief Engineer depending upon agreements between Air Force Life Cycle Management Center (AFLCMC) Program Offices and the SCMG.

Figure 3. TPS Software Deficiency Correction Process.



5. Project Submission Requirements Process.

5.1. Data Call. First of each year (January) a requirements data call is sent out by Strategic Sourcing Program for upcoming FY TPS software maintenance requirements. Projects shall be developed, input, reviewed, signed and submitted to Strategic Sourcing Program before EOM February for organic and by 15 March for contract TPS. Upon submittal of projects to Strategic Sourcing Program (2.4 Status), recommend adding applicable stakeholders to MPST distribution list for project updates. In addition, a notification is posted to the project Message Board notifying Strategic Sourcing Program project is ready for validation. Follow-up may be accomplished by phone or email. In-cycle TPS projects do not have a ranking process. When deciding how to proportionally distribute available funds, if not fully funded, other factors such as mission impact may be considered.

5.2. Out of Cycle (OoC). Project submittals established after the cutoff date (EOM February for organic and by 15 March for contract TPS) or during the year of execution will be submitted as an OoC requirement. Funding from non-critical in cycle projects may be moved to critical OoC projects depending on the circumstance. Validation status of OoC shall be determined by Strategic Sourcing Program within 15 working days of project submission**.**

5.3. Submitting projects. The originator/RG (engineers, program managers, equipment specialists, or supply plannersubmits candidate requirements once a potential deficiency in software is identified. TPS software maintenance engineering projects require specific documentation identified to the TPS deficiency at the NSN level (typically a DR, IR, or SWDR) by weapon system. All projects must meet program scope and satisfy submittal requirements prior to approval. Project requirements not entered in their entirety or documented properly within MP&E, MPST, and/or any other applicable system database; will result in project disapproval.

5.3.1. Non-stock listed items falling within CSAG-S criteria or within the scope of its management must have a stock listing action initiated before project submission.

5.3.2. Planning shall begin as soon as the project need is identified. To ensure timely development and implementation of projects, recommend a collaborative Integrated Product Team (IPT) be formed to include; but not limited to, SPO and Supply Chain Management Group (SCMG) members consisting of Program Managers (PM), Engineers (EN), Equipment Specialists (ES), Supply Support, Contracting Officers, Financial Specialists, Software Organization (SWO), and maintenance POCs as required. Minutes from planning meeting, critical decisions, timelines, list of attendees, and correspondence via e-mail may be uploaded to MPST file repository for historical and supporting documentation purposes.

5.4. Software Maintenance Source Support. As projects are submitted each year, a sound basis of estimate is necessary to lend credibility to the definition, validation and justification of the resources needed for funding. Basis of estimates from Software Maintenance sources should include at a minimum, DR, IR, Man Hours per DR/IR, Labor Rate and an estimate for materials (to include itemized list), if applicable. As deficiencies are resolved throughout the year, software support personnel will be asked to provide monthly execution status to include hours applied, DR opened/closed, dollars expensed throughout project completion. Expense dollars reported should reflect the billed amounts.

5.5. MP&E System Requirements. The MP&E system provides information used to posture depot maintenance facilities and the development of the CSAG-MD budget. MP&E provides a common system for negotiating maintenance cost, schedules and the allocation of maintenance workload among organic inter-service and commercial repair sources. MP&E is a depot level maintenance requirements and program management system and Authoritative Source (AS) for Program Control Numbers (PCNs). AFMCMAN 20-102, Chapter, 9 provides MP&E policy, procedures and responsibilities for developing long range customer workload requirements specific to software Repair Group Category (RGC) - S (Software) requirements. Repair Program Managers (RPM) will request a PCN and file maintain requirements in MP&E for all assigned TPS workload. This requirement will be utilized by the SWEG organization for planning and initiating projected FY workload requirements to be approved in the R2D2 process.

5.5.1. R2D2. AFSC/LGP is the point of contact for all AFSC Other Funded Customer Orders (formally known as Direct Cite) submitted to the R2D2 process. Upon receipt of the Annual R2D2 Enterprise Access Management Service (EAMS) takser, AFSC/LGP will issue a Data Call to their respective Organizational Workflows and RPM, ensuring all participants are notified and provided detailed instructions to complete file maintenance in the MP&E system IAW file maintenance suspense dates listed in the EAMS/Data Call. The primary purpose of the R2D2 process is to ensure AFMC has the required manpower and capacity budgeted and in place to meet peacetime depot maintenance missions in the most economical and effective manner. R2D2 provides for centralized planning, analysis and control of organic depot maintenance resources and workloads managed by the Maintenance Complexes/Wings in accordance with AFI 21-101 AFMCSUP, *Aircraft and Equipment Maintenance Management*, and HQAFMC/A4D and AFSC R2D2 supplemental annual guidance.

5.5.2. MP&E uses the AFMC/FMR Sales Rate Memo (provided July of each calendar year) to file maintain requirements based on the RGC sales rates for current year and seven out-years. For RGC-S workload, a requirement is: the hourly rate times total hours equals the total dollars. The RPM must file maintain this requirement in MP&E. All UUT requirements MUST be file maintained in MP&E and MPST for approval and planning of requirements. The hours file maintained in MP&E MUST be used when projecting hours in MPST.

5.5.3. AFSC/LGP is the Source of Supply (SOS) for purposes of R2D2 and will validate all file maintenance prior to the posture planning process where the inflation factors will be applied.

5.6. MPST Requirements. MPST is a maintenance engineering tracking tool used to submit, review, validate and approve funds for Sustaining Engineering CSAG-S/GSD (General Support Division) and TPS soft- ware maintenance engineering requirements. Within the tool, projects flow through a tiered-process documenting status from input, review, validation, and funding. Each project must include all required MPST elements completed in its entirety prior to project submittal. **Table 4** lists all required elements required for TPS projects.

5.6.1. Methodology for Calculating Requirements. In the MPST project form, show the methodology used to calculate your requirements. The man-hour requirements for organic and cost for contractor must be based on a methodology, documented, and retained by the MPST POC for a period of one year after completion of the task. Prior year RGC-S rates or upcoming year Planning Resource Control Center (RCC) Shop rates may be used when calculating TPS organic software maintenance requirements (upcoming year RGC-S rates are not published until July each year). RGC-S rates include direct costs (to include material), overhead, and G&A. RGC-S software sales rates are an average of the "rolled up" RCC rates across each of the Complex SWEG Software shops and may not be the rate that is charged to the customer. RCC Shop rates may be used for estimating project total dollars when the RCC rate is much higher than the RGC-S rate for the upcoming year. In this case if material costs are required, projected direct material costs must be added to the total dollars when using the RCC shop rate.

5.6.2. Planning RCC rates for the upcoming year are usually finalized in early February each year and may be more realistic than using the RGC-S rate for your Complex. If you elect to use the RCC rate and direct material is required, you must include an itemized list of material with total dollars estimated (using current or past year actuals) with your project submission. Material cost documentation must be provided and uploaded in MPST file repository to support the use of the RCC rate + material cost versus the RGC-S rate. Provide justification for using the RCC rates in the body of your project; usually the Delta section of the Project Info Form. For MP&E, only hours must match MPST. Hours and Rates used in MPST may not match MP&E if RCC + material costs are used. Methodology for calculating a TPS organic requirement may include the following:

5.6.2.1. **(IR + DR)** * Man Hours) * Prior Year RGC-S) = Total TPS organic requirement $$.

5.6.2.2. **(IR + DR)** * Man Hours) * Upcoming Year RCC Rate + Material Costs) = Total TPS organic requirement $$.

5.6.2.3. Basis of estimate from Software Maintenance sources should include at a minimum, DR, IR, Man Hours per DR/IR, Labor Rate and an itemized list of materials if Composite Rate is not used. Historical trending, assumptions, quotes, etc. used to support your calculations should be uploaded to the MPST file repository.

5.6.3. FY Funding dollar projections are located in the Funding section of the MPST Project Info Form. Funding dollars are input in millions ($M) by FY. These inputs are required to reflect the funding projections required for your project and should be consistent with the funding dollars planned and file maintained in R2D2.

5.6.4. Projects not meeting MP&E requirements with an established PCN assigned and associated data elements satisfactorily completed prior to establishing an MPST project will not be approved for CSAG-S funding and no work shall be accomplished.

5.6.5. For Organic TPS, the hours file maintained in MP&E must be used when calculating TPS funding dollars in MPST. RGs must be able to provide supporting documentation for dollars and rates and other costs used. A basis of estimate, Request for Quote (RFQ) reflecting the hours, rates, etc., used to calculate your requirements should be provided and uploaded to MPST file repository. After inflation factor is applied in MP&E (NLT 31

March each year), inflated RGC-S rates may not match what was submitted in MPST. RGs will reconcile MPST with the adjusted rates and FY funding dollars.

5.6.6. For Contract TPS software maintenance engineering requirements, a basis of estimate/RFQ should be used to estimate project dollars. Estimates should include hours, rates, and estimated DR, if known, for DMISA TPS software maintenance engineering requirements, a basis of estimate/RFQ should be used to estimate project dollars. Estimates should include hours, rates, and estimated DR if known.

5.7. Execution Schedule. The project execution schedule is an internal MPST project form located under the Financial Data tab in the drop down. The estimated execution schedule should be forecasted in MPST NLT 15 May. All projects shall have a funds execution schedule detailing each phase of the TPS software maintenance engineering project significant developments. Funds execution schedule includes financial commitment, obligation, and expense milestones. The forecast and get well commitment and obligation dates input to MPST execution schedule will be used as the baseline to monitor project execution. Commitment and obligation dates are used for forecasting requirements. Due to the nature of software maintenance, forecasting expenditure dates for TPS is not feasible as software maintenance is only performed as discrepancies arise.

5.7.1. Organic TPS. As a goal, AFMC206 acceptance (commitment and obligation) should be accomplished by 31 Dec. In those cases where incremental funding is applied, Forecast, Get Well, and Actual/Completion dates will be provided in MPST.

5.7.2. Contract TPS. Upload timelines to include forecast, Get Well, Actual/Completion dates for Statement of Work (SOW), ROMs, Funds Committed, and Obligated in MPST. These dates need to be coordinated through PK. **Table 3** Illustrates the data required for tracking Contract TPS execution.

5.7.3. DMISA TPS. Upload timelines to include forecast, Get Well, Actual/Completion dates for SOW, ROMs, Funds Committed, and Obligated in MPST.

Table 3. Contract TPS Project Tracking Data Requirements.

Project Initial Documentation Requirements	IPT Meeting Minutes, list of attendees
	Basis of Estimate (BOE)/Rough Order of Magnitude (ROM)
	PWS/SOW completion date
	Requirements Approval Document completion date
	PR Number
	PR Package completion date
Estimated dates shall be updated each month until award.	Estimated or Actual PR commitment date
	Estimated or Actual contract award/obligation date
	Contract Number
	Contract award amount in dollars
Monthly Reporting Requirements	Status report to include IR, DR opened and closed and total monthly dollars expensed.
Final Contract Completion Report Requirements	Contract award date
(Due in year which project contract completed)	Start and end of Repair Service contract, indicate if contract is multiple year contract
	Final total project cost at completion
	Total number of discrepancies opened and closed

5.8. Required Elements. All TPS software maintenance engineering project submittals will be input and processed in MPST to include all applicable supporting documentation. **Table 4** below reflects the required elements for both contract and organic TPS. Each of these elements will be input directly into the MPST project form. For contract TPS, see **Table 3** for additional data requirements.

Table 4. MPST Required Elements.

Required elements within MPST projects include:
Each project identified by a unique numeric project identifier.
Program Control Number (PCN). For DMISA inter-service PCNs, ensure Source of Repair (SOR) has been designated as DA (Department of Army), DN (Department of Navy) or CG (Coast Guard).
NSNs supported by the software change. A full list must be added or uploaded in MPST NIIN field.
Number of deficiencies estimated to be considered - must be justified and documented in MPST.
Estimated # DR considered and corrected for the project.
Estimated number of man-hours per software deficiency correction.
Project cost (labor, materials, and other direct costs) per FY. Upload Software Maintenance source Basis of Estimate to file repository supporting your project total costs (rate, hours, DR).
Estimated project cost must be substantiated with auditable documentation. Calculation for funding must be shown in narrative section of MPST (i.e. Hours per SWDR x # of SWDR x labor rate] + direct costs = proposed project cost) or (Man Hours x labor rate).
Signatures. At a minimum, approval by the EN Flight Chief, Supply Chain Manager and Group DoE. MAJCOM, Weapon System Program Manager and/or Chief Engineer as required.
FY Funding $$ projections. Narrate to Deltas over $100K or any variance (+ or – 25%).
Purpose as to why the requirement is needed. If Program Management Directive (PMD) directed, provide PMD #.
Description to include deliverables, technical information, specific software to be updated, and explanation of potential changes.
Justification/Impact - State the impact on the warfighter's (MAJCOMs) ability to perform its mission. (The loss of expertise and shop personnel is not adequate justification)
Execution Forecast. For Contract and DMISA only. Coordinated Forecast Commitment and Obligation dates must be input to execution schedule. Initial projections must be updated to MPST NLT 15 May.

5.9. Signature Process. Prior to submitting to Strategic Sourcing Program for validation, all projects must go through the signature review process. A required list of signatures is provided in **Table 5**.

5.9.1. Panel Review. Typically projects will be reviewed through the panel review process. The panel may consist of the engineer, flight supervisor, program manager, equipment specialist and DoE and is the interface between SCMG engineers and Strategic Sourcing Program. Each Group's project review panel will review projects and supporting documentation to ensure requirements meet scope, validate requirement calculations, and verify MP&E and MPST requirements are accurate and complete. The panel shall formally evaluate, validate, and approve projects. Once all requirements have been met, project status should be changed to "2.4 Submitted for Program Office Review" and a consolidated list of the Groups' projects will be sent via e-mail to Strategic Sourcing Program workflow, **429SCMS.GUMD.SEPOSupport@us.af.mil**, prior to Data Call cutoff date.

5.9.2. Clarification of Signatures.

5.9.2.1. SCMS/EN Flight Chief (Required) – The flight chief signature represents the 1st level of review for the project, ensures completeness to include supporting documentation and has been vetted through the IPT level functional and technical peer review is complete. The Flight Chief will also be responsible for moving the project to certain statuses throughout project coordination. The corresponding MPST status is 1.1. Submitted for Flight Chief Review/Sign.

5.9.2.2. DoE (Required) – The DoE signature represents the approval level for the project. Prior to DoE signature, the project has gone through SCMG/EN FLT chief review. DoE certifies project scope and justification aligns with sustaining engineering efforts. In addition to the signature request, the DoE will also mark a DoE Data Call box on the MPST page after approval. Projects amended for cost increase greater than 10% or supporting information changes will require re-certification of signatures. This is done at the 1.2 status.

5.9.2.3. SCMS/CC/CL/DD (Required) – The CC/CL/DD signature represents coordination and chain of command awareness after FLT Chief and DoE levels have been obtained. Project is within technical scope and prioritized appropriately across the Groups. Projects amended for cost increase greater than 10% or supporting information changes will require re-certification signatures. The corresponding MPST status is 1.3 Submitted for SCMS Review/Sign.

5.9.2.4. System Program Manager (SPM)/Chief Engineer (Optional) – The SPM/Chief Engineer signature represents coordination and chain of command awareness. As the designated Operational Safety, Suitability, and Effectiveness (OSS&E) Office of Primary Responsibility (OPR), the Chief Engineer has knowledge of any efforts on the System Program Office side that would represent modifications planned or other flight requirement changes impacted by higher HQ USAF or Department of Transportation or Federal Aviation Administration (FAA). An example would be verifying that FAA does not plan to eliminate acceptable flight transponder modes within the same timeframe that the project is underway. The SPM/Chief Engineer acknowledges that the Weapon System will be in service beyond timelines for this project. This signature is not an approval. The Weapon System level of involvement may vary depending on

cognizant engineer relationships between Supply Chains and Program Offices. The SPM/Chief Engineer may expect updates on project status. SPM/Chief Engineer implies assistance with MAJCOM and flight testing coordination. SPM/Chief Engineer signature is requested after SCMG signatures are received. SPM/Chief Engineer signature may be optional for TPS projects, but required for CSAG-S/GSD projects. The corresponding MPST status is 1.4 Submitted to Lead SPM Review/Sign.

5.9.2.5. MAJCOM (Optional) – The MAJCOM representative or representatives are coordination that the weapons system plans are to have the weapon system remain in service while the sustaining engineering project is underway against a NSN. MAJCOM signature is not implying project approval, but is often requested later in project development after supply chain and weapon system signatures. MAJCOM signature may be optional for TPS, but required for CSAG-S/GSD projects. The corresponding MPST status is 1.4. Submitted to Lead MAJCOM Review/Sign.

Table 5. Project Panel Coordination and Signature Requirements.

	1) SCMS Engineering Flight Chief
	2) SCMG DoE
	3) SCM Squadron CC/CL/DD
	4) Resource Advisor
	5) 448 SCMW FM

5.10. Supporting Documentation. Retention of documentation and products to support the method used to calculate your requirements should be held for at least one year after completion of project. Data collected to justify requirements through deficiency reports like JDRS, Organic Change Proposal (OCP), Material Improvement Plan (MIP) and/or any other statistical method used to calculate the requirement from historical trends should be retained and/or uploaded to MPST file repository. All documentation (basis of estimate, ROMs, RFQs, etc.) for estimated rates, man-hours, DR, IR, additional costs, key assumptions, specific details, decision makers' names and signatures and dates of decisions should be retained to support your requirement and uploaded to MPST file repository. Other documents received from a vendor or maintenance source may be included if they are necessary to support requirement calculations.

5.11. Project Validation. Strategic Sourcing Program will conduct a review of Group DoE approved submitted projects ready for Wing-level validation to ensure requirements are accurately defined, project fields completed within applicable tools/systems, and proper documentation provided. During validation process if additional information is required, Strategic Sourcing Program will notify RGs via message board, phone or e-mail until all required information is received. Projects requiring additional information will be moved to 3.0 Additional Information requested by Program Office. Projects not meeting required criteria will be disapproved and moved to 3.8 status.

5.11.1. Strategic Sourcing Program submits list of validated Wing requirements to HQ AFMC/A4UE for coordination and approval of FY approved projects list.

5.11.2. Strategic Sourcing Program submits validated approved requirements to the 448 SCMW/FM.

5.12. Funding Projects.

5.12.1. 448 SCMW/FM submits requirements to HQ AFMC/FM for budget submissions. 448 SCMW/FM notifies Strategic Sourcing Program of upcoming FY funding level (typically October–November timeframe).

5.12.2. Strategic Sourcing Program will establish a comprehensive and dynamic list of funded and unfunded projects. This list will be published on the 429 SCMS/GUMD SharePoint® monthly starting in the October – November timeframe and updated as necessary. Both in-cycle and out-of-cycle projects will be listed as well as those pushed forward or pulled back. Projects on hold but validated will also be included on the list.

5.12.3. HQ AFMC/FMR provides funding authority to the 448 SCMW/FM.

5.12.4. Per Strategic Sourcing Program direction, 448 SCMW/FM provides funding authority and distribution to each 448 SCMW/FMA-OL. 448 SCMW/FMA –OLs will move project to 3.7 once funding has been distributed.

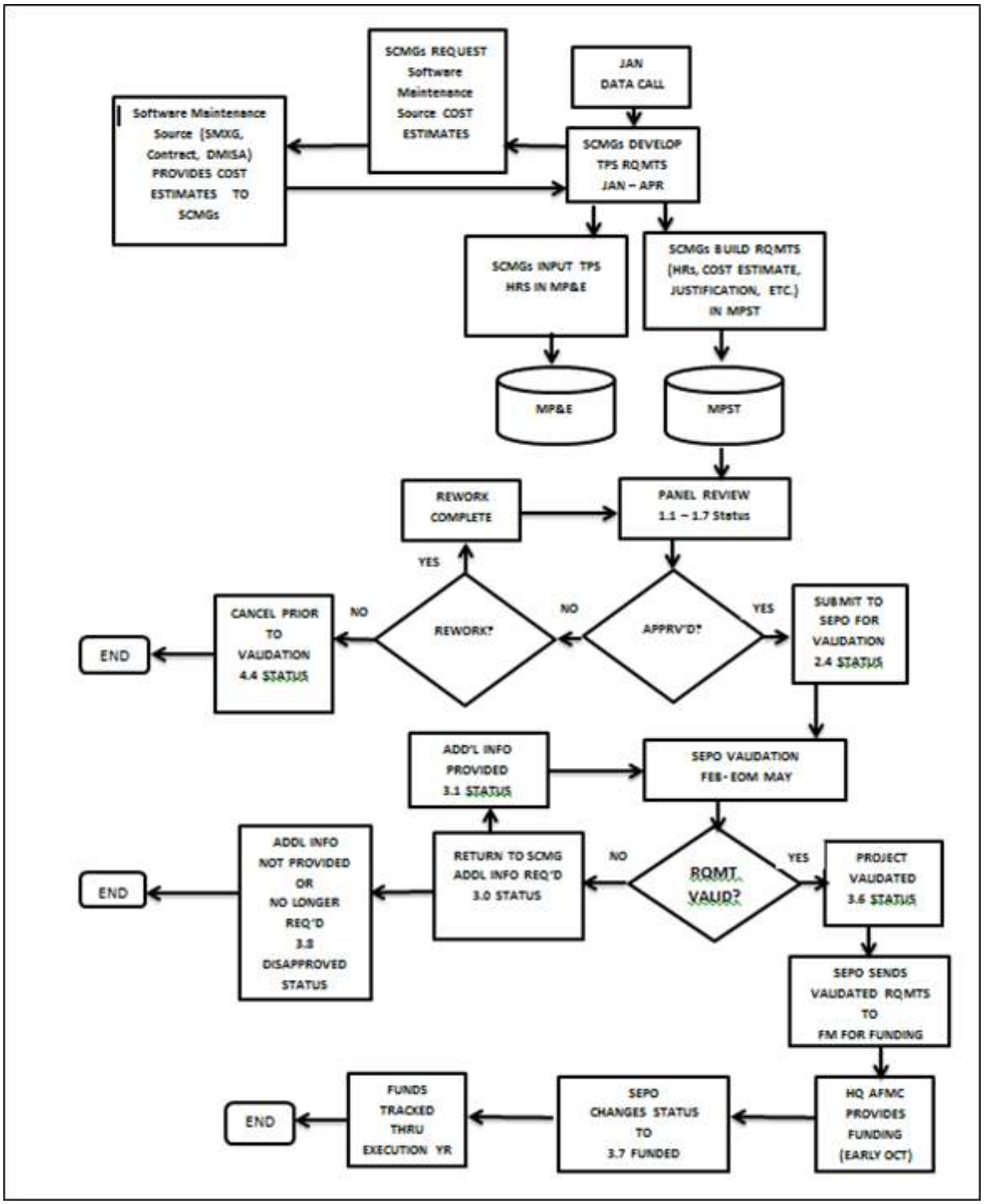
5.12.5. 448 SCMW/FMA-OL ensures project funds are loaded onto respective projects in financial systems for execution.

5.12.6. For organic, after receipt of Project Order AFMC181 from resource advisor, initiate AFMC206 to establish PCN for induction and funds execution as early as 1 October each year. For contract TPS projects, the RG initiates the PR and prepares applicable documentation for the procurement package. For policy guidance on the generation and processing of Purchase Requests, see AFMCI 20-102, *Requirements Definition and Purchase Instrument Development*.

5.12.7. The RG generates applicable documentation actions for TPS contracts, i.e. Requirements Approval Document (RAD), Performance Work Statement (PWS)/SOW, Justification & Approval (J&A), etc.

5.12.8. The RG cannot fund TDY from project funding. Government travel in support of validated TPS maintenance projects are funded separately under the appropriate EEIC 409. Government travel may not be funded under TPS EEICs 56001 (contract), 54001 (organic) or 54002 (DMISA).

Figure 4. CSAG-S TPS Sustaining Engineering Project Process.



6. Execution Year.

6.1. Funds Execution.

6.1.1. All TPS funds are used for investigation, analysis, correction of deficiencies discovered during year of funding and/or pending TPS maintenance testing issues. Correction and investigations may have to be deferred to the next funding cycle due to lack of funding, or additional SWDR being generated (more than predicted). Funding is valid for open JON requirements. Our goal is to have AFMC Forms 206 accepted (committed and obligated) by 31 Dec. Projects unable to meet this goal will be at the discretion of the DoE and Strategic Sourcing Program recommendation. Contract TPS commitment and obligation dates may vary by project but must occur prior to end of Fiscal Year. Excess or surplus funds not expected to be obligated (placed on AFMC206) during execution year must be returned as soon as identified and prior to end of year closeout. Where funds have been obligated but the anticipated DR have not generated, funding must be returned prior to end of year closeout. Generated requirements (DR, IR, etc.) that will not be completed by end of fiscal year must be validated to ensure funding for open JONs (AFMC206, Production #s, and T- jobs) are retained.

6.1.2. 448 SCMW/FMA-OL Resource Advisors monitor/track funds execution in MFT, FM Suite and J025A. Resource advisors work closely with RGs to ensure project funding is available. Resource Advisors shall communicate any excess, funding shortfalls or potential over-runs to RGs and Strategic Sourcing Program.

6.1.3. Expense reporting is required monthly and will include at a minimum, AFMC206 (JON, Production #, T-job #) document numbers and closure date, # IR, DR opened/closed, hours worked, dollars expensed, funding shortfalls and excess to requirements. Expense reporting will be provided by Software Maintenance Support POCs to RGs. RGs will provide data via 429 SCMS/GUMD SharePoint® until MPST is developed.

6.2. Monthly DoE Project Status Meetings. These meetings will be organized by the Strategic Sourcing Program at the discretion of site DoEs. Status updates should be provided no later than the second Tuesday of each month, unless other arrangements are mutually agreed to by site DoE and Strategic Sourcing Program.

6.2.1. Software Maintenance sources (SWEG, DMISA or Contract) report periodic TPS maintenance development status, i.e., IR, DR, hours and dollars expensed, funding shortfalls, surplus funds, and open JON funding requirements to RGs.

6.2.2. The RG validates software maintenance source reports for SWDR complete, labor hours worked, and funds expensed.

6.2.3. The RG updates MPST to reflect latest status of project execution. At a minimum, for organic/contract/DMISA TPS the following information will be requested via Strategic Sourcing Program spreadsheet until MPST tool is enhanced: total hours and dollars planned, number of investigations opened and closed, number of DR opened and closed, 206 document numbers (Production #s, T-Job, JON) with close date, monthly total funds expensed and monthly total hours spent by project. Additional information may be requested as TPS project status reporting is further developed. Strategic Sourcing Program spreadsheet will be saved to 429 SCMS/GUMD SharePoint® for monthly update reporting.

6.2.4. The RG attends project status meetings to update leadership on project execution and workload status. If the RG does not have the right knowledge to address a specific issue as a projects matures, an IPT member, i.e., program manager, financial advisor, contracting specialist, etc., may be called upon to provide critical knowledge of the project as it moves within its life cycle; and should also be notified to attend meetings.

6.2.5. Project reporting is used to track the progress of the project to ensure funding is obligated prior to end of FY for contract TPS workload and ensure obligation funding will be fully expensed and utilized for execution year organic TPS project workload. All the funded projects will be monitored by the RG and monthly updates will be provided to Strategic Sourcing Program via MPST and/or 429 SCMS/GUMD SharePoint®. RGs and/or SCMS Flight Chiefs will brief latest status at DoE monthly meetings.

6.3. File Maintenance. All project submittals that are validated and funded shall require periodic status reporting and tracking in MPST and/or 429 SCMS/GUMD SharePoint®. Each RG shall ensure periodic file maintenance as required to reflect project developmental progress via MPST. Updates will be provided for each funded project of that fiscal year until that project is completed and all deliverables have been received by the government. For contract TPS, a list of required data tracking requirements from initiation through project completion and closeout have been provided in **Table 3** All tracking, status updates and supporting documentation shall be uploaded in the applicable systems to include the MPST document repository.

7. Budgeting Information.

7.1. Background. CSAG-S TPS (4930 Appropriation, Fund Code-64, EEICs-54001 Organic, 56000 for Contract and 54002 for DMISA) and associated engineering activities support restoration of Budget Code 8 NSNs. This section applies to all AFMC controlled organizations, which support CSAG-S/TPS items. Budgeting seeks to establish baseline funding levels for future requirements over a specific period of time, usually referenced by FY. The President's Budget is a consolidation of all program budgets submitted during a Program Objective Memorandum (POM) cycle. The POM is a major document in the Planning, Programming, Budgeting and Execution (PPBE) process, and the basis for the component budget estimates. The CSAG-S/TPS Sustaining Engineering (SE) requirement is developed from individual projects and entered into the Budget Execution Submission (BES) with inflationary factors applied to out years. The BES is consolidated by Air Staff, which ultimately becomes part of the POM. HQ AFMC/FMR issues the budget figures obtained from the Program Budget Review (PBR) down to the AFSC/FM, and then AFSC/FM issues a funding document to the 448 SCMW/FM where the funds are executed.

7.2. POM Requirements Input. Requirements are collected by Strategic Sourcing Program on an annual basis to facilitate the budget process, allowing lead-time for requirement identification/submission under the appropriate budget system (BES, PBR, POM, etc.). At the time requirements are identified, they are considered a current year requirement even though they are assigned to a FY based on the projected year of execution (budget year).

8. Funding Allocation.

8.1. Fund Disbursements. Funds are distributed to the Wing for site disbursement. AFSC/FM issues an Annual Operating Budget (AOB) to 448 SCMW/FM on or about 1 October. The 448 SCMW/FM AOB sets the funding limitations by which Group financial staff will allocate funds to the requirements owning organizations. All CSAG-S funds for TPS must be obligated within the year of execution. Funding must be loaded onto the individual validated projects so that execution and tracking may begin. CSAG-S financial POCs shall load the approved authority to each project. 448 SCMW/FMA-OL POCs ensure funds are loaded in CRIS-BQ, MFT and J025A. This authority will display on the execution summary reports and should match the funding loaded in BQ and/or applicable financial tools.

8.2. Adjustments. Adjustments related to CSAG-S/TPS software maintenance engineering projects are classified into the following two categories listed in **Table 6** Adjustments typically are associated with fallout, funding shortfalls and new requirements, which occur periodically throughout the life cycle of each project. Sites must request authorization from Strategic Sourcing Program to realign funding between validated projects during the year of execution to meet mission requirements and schedules as they are identified. RGs should also inform Strategic Sourcing Program and 448 SCMW/FMA through message board of project shortfalls. Strategic Sourcing Program may require DoE notification where applicable. Allocation to OoC projects can only be accomplished after the Strategic Sourcing Program validated status has been applied and approved. For both contract and organic projects, at no time can the project allocation amount exceed the funding authorized by 448 SCMW/FM. If a prior year organic or contract TPS project requires additional funds after year of execution, 448 SCMW/FM must identify funds are available in FY required, otherwise a current FY project must be submitted in accordance with TPS project submission requirements.

Table 6. Adjustment Categories.

Current Year	Any under/over execution of contracts or other source documents (i.e., AFMC206) after initial distribution of funding can be classified as a current year adjustment if the project is using current year funding.
Out of Cycle	Out-of-cycle adjustments exist whenever a new requirement develops and it is deemed critical funding be sought to accomplish the mission objectives within the current funding cycle with existing or excess funds.

8.3. Surplus Funds. Funds generated from a reduction in a project's original requirement, cancellation of a project or funds from projects unable to execute in accordance with established target dates will be classified as surplus or excess. Software Maintenance source POCs will notify RGs when funded hours are excess to current requirements or projected workload. In turn, RGs will notify Strategic Sourcing Program Office and 448 SCMW/FMA via MPST message board requesting realignment of surplus dollars/hours along with a written explanation for return of funds. Strategic Sourcing Program Office may require DoE notification where applicable. All excess funds shall be de-committed/de- obligated from the contract upon earliest notice and realigned to center accounts. Group DoEs shall ensure MPST

will be updated to reflect the change. Under the direction of 448 SCMW/FM, surplus funds will be dispersed according to Group discretion across the Wing unless otherwise directed by 448 SCMW/EN. **Note**: After the requirement is complete, any excess funds CANNOT be applied to any other requirement without DoE, Technical Director and 448 SCMW/FM approval and a courtesy copy to Strategic Sourcing Program site representative. Additionally, current FY funds cannot be applied to any other requirements outside the FY of funding.

9. Project Closeout. In order to maintain accountability and traceability of each project from beginning to end, project close-out actions must be accomplished. For Organic TPS, outstanding JONs (AFMC Forms 206, Production #s, T-jobs, etc.) should be closed after completion of work (no outstanding tasks to be completed, no additional hours/dollars required to complete tasks). For Organic, Contract and DMISA projects, Software Maintenance source POCs will provide a close out status report to RGs for each software maintenance engineering TPS project. At a minimum, the following information will be provided: total hours and dollars charged to the project, total DR, IR opened and closed, any remaining hours and dollars to be returned, indicating that no other open tasks are remaining which need to be charged. RGs will upload to MPST and notify 448 SCMW/FMA-OL via message board. 448 SCMW/FMA-OL will ensure all uncharged dollars will be de-committed/de-obligated and provide project close out date (date all 206 have been closed, funds de-committed and de-obligated).

9.1. TPS Maintenance Project Closing Action (Organic/Contract/DMISA).

9.1.1. If applicable, RGs will determine with assistance from maintenance source POC, when a project and all associated IR, DR actions are complete and will initiate workload closing action.

9.1.2. Enter the project outcome in MPST.

9.1.3. If applicable, enter number of DR, IR opened and closed.

9.1.4. Enter cost/labor hours expended.

9.1.5. Upload close-out report to MPST file repository.

9.1.6. Contract TPS Data Tracking Requirements. Upon contract completion and government acceptance, a final report for each project shall be uploaded in the MPST to include government acceptance of deliverables, contract close-out and documentation showing final dollar award. Reference **Table 3** for Contract TPS Data Tracking requirements.

9.2. Project Cancellation.

9.2.1. Projects may be cancelled at any time.

9.2.2. Projects cancelled after validation must follow the program designated cancellation process with Group DoE coordination facilitated by MPST tool.

9.2.3. MPST message board must be updated to state the reason the project was cancelled.

10. Support Systems.

10.1. USAF Standard Base Level General Accounting and Finance System (H069/BQ). Used to load and execute cost authority. CRIS is the interface for BQ.

10.2. Commanders Resource Information System (CRIS). Extracts financial data stored in BQ. It allows customized reporting of financial transactions that can be used to manage CA assigned and their impact. Currently, CRIS remains the primary system from which MPST CA updates are drawn. A regular reconciliation must be accomplished to keep the execution records accurate and in agreement with CRIS.

10.3. Defense Industrial Financial Management System (DIFMS). Used to track both production hours and cost, for organic only. It is the Air Force official financial system of record for organic depot maintenance.

10.4. Job Order Production Master System (G004L). Used to track production hours worked against a JON (AFMC206).

10.5. Joint Deficiency Reporting System (JDRS). Engineering system used for Deficiency Reporting, Investigation and Resolution (DRI&R), including software.

10.6. Automated Project Order (APO) system (J025A). where organic workload documents are maintained and funded with CA. APO automates the AFMC181 process and coordination, and provides an interface with the DIFMS and the G004L systems. The emphasis of this system is to provide access for the status of documents as approved, rejected, or cancelled; for the extraction of data for summarizations; access to historical data; and reports.

10.6.1. Ceiling represents the amount of authority loaded when AFMC181 is entered into JO25A. The ceiling sets the limit to which financial transactions can be processed.

10.6.2. In-Process represents the amount of funding on an AFMC206 that has been entered into the G004L system but has not been accepted by DIFMS.

10.6.3. Accepted is the term used to identify funding that has been accepted by DIFMS, which now authorizes the planning, and induction of specific workloads.

10.6.4. Reports/Reporting. All of the reports used to report CA can be found under the Main Menu option - “Incoming”.

10.6.5. Cost Authority (CA) available balance. Anytime you need to view a summary level report of funding status, CA Balance report should be your choice. The CA Balance report shows you a balance for each Maintenance Priority Code (MPC) as it progresses through each stage of accounting based on report dates. CSAG-S (organic DMAG) Engineering Support is MPC code 3R02.

10.7. Maintenance Planning & Execution (MP&E). Provides a single source of depot level requirements for the Air Logistics Complexes (ALCs) and other Air Force Logistics facilities. It is the official Authoritative Source (AS)/system of record for establishing and assigning PCNs. Reference AFMCMAN 20-102.

10.7.1. MP&E information is used to posture depot maintenance facilities and the development of the Consolidated Sustainment Activity Group – Maintenance Division (CSAG- MD) budget.

10.7.2. MP&E provides oversight of maintenance program planning and management functionality for major and secondary items developing long range customer workload requirements for: Aircraft, Missiles, Engines, Other Major End Item (OMEI), Exchangeables, Area Base Maintenance, Manufacture, Software, and Storage. MP&E portrays customer requirements, programs, and workload identity through RGC, Source of Repair and reimbursement Customer.

10.7.3. MP&E provides information to support the Requirements Review Depot Determination (R2D2) Process, Core Review, Base Realignment and Closure (BRAC), and War Surge.

10.8. Maintenance Engineering Project Sustainment Tool (MPST). Government owned/ web-based program used to initiate, track, validate, administer, and report on the accountability of CSAG-S/GSD/TPS engineering program projects. The system is updated on regular intervals with financial data downloaded from CRIS. Ad hoc reports can be/have been developed to help manage the CSAG-S/GSD/TPS programs. It provides online assistance for completing the required submittal form. A user's manual is posted on the home page. MPST Tool can be accessed at the following web site **https://msdis.us.af.mil/msd_engineering/home.asp**.

10.9. On Line Report Viewer (OLRV). Funds Initiator Report is the central processing point for all organic workloads. This report attempts to provide information on the status of funds throughout the life of the AFMC181, Work Request (funds authorization) and the AFMC206, Temporary Work Request. When the AFMC181 is processed it will load funds authorized and funds available to bill. When the AFMC206 is processed an amount will be entered into the Customer Order Numbers (CONs) required column which should be the amount on the AFMC206. Any excess/short amount will be calculated from the authorized, less CONs required. It should also be noted that all Job Order Numbers (JONs) will be separated out and could cause multiple lines with the same dollar amounts since the billing is keying on the Fund Classification Reference Number (FCRN) level. If a balance remains at the end of a quarter then the Sponsor Order Number (SPON) will change to reflect the new Purchase Order Number (PON) in most cases.

11. Records. Beginning FY06, all TPS records contained within the web application MPST are accessible by going to hyperlink **https://msdis.us.af.mil/msd_engineering/home.asp**. The web server and data are maintained at Hill AFB. Funding for the application is managed at Hill AFB. All data entered into the application is stored indefinitely.

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Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 21-101, *Aircraft and Equipment Maintenance Management,* 15 January 2020

AFI 21-101 AFMCSUP, *Aircraft and Equipment Maintenance Management,* 09 November 2020

AFI 33-322, *Records Management and Information Governance Program,* 23 March 2020

AFI 65-118, *Air Force Purchase military Interdepartmental Purchase Request (MIPR),* 7 April 2020

AFMAN 63-143, *Centralized Asset Management Procedures (CAM),* 18 December 2020

AFMAN 65-605, Volume 1, *Budget Management for Operations,* 24 October 2018

AFMCMAN 20-102, *Maintenance Planning and Execution System (MP&E) (D363)*, 7 January 2019

OPNAVINST 4790.14B, *Joint Depot Maintenance Program,* 14 February 2013

T.O. 00-35D-54, *Technical Manual: USAF Deficiency Reporting, Investigation, and Resolution*, 1 September 2015

Prescribed Forms

None

Adopted Forms

Air Force Form 847, *Recommendation for Change of Publication*

AFMC Form 206, *Temporary Work Request*

DD Form 448, *Military Interdepartmental Purchase Request*

AFMC181, *Project Order*

Abbreviations and Acronyms

AF—Air Force

AFI—Air Force Instruction

AFLCMC—Air Force Life Cycle Management Center

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFPD—Air Force Policy Directive

AFSC—Air Force Supply Chain

ALC—Air Logistic Complex

ALO—Accounting Liaison Office

AOB—Annual Operating Budget

APO—Automated Project Order

ATE—Automatic Test Equipment

ATS—Automatic Test System

BES—Budget Execution Submission

BOE—Basis of Estimate

CAM—Centralized Asset Management

CAV—Contract Asset Visibility

CC—Combatant Commander

CCB—Configuration Control Board

CG—Coast Guard

CL—Civilian Leader

CON—Customer Order Number

CIP—Capital Investment Program

CPIN—Computer Program Identification Number

CRIS—Commanders Resource Information System

CSAG—Consolidated Sustainment Activity Group

CSAG-S—Consolidated Sustainment Activity Group - Supply

CSAG-MD—Consolidated Sustainment Activity Group – Maintenance Division

CSS—Control and Support Software

DA—Department of Army

DD—Defense Department

DD—Deputy Director

DIFMS—Defense Industrial Financial Management System

DMISA—Depot Maintenance Inter-service Support Agreement

DN—Department of Navy

DOD—Department of Defense

DoE—Director of Engineering

DPEM—Depot Purchased Equipment Maintenance

DR—Deficiency Report

DRI&R—Deficiency Reporting, Investigation & Resolution

DWCF—Defense Working Capital Fund

EAMS—Enterprise Access Management Service

EEIC—Element of Expense Investment Code

EN—Engineering

EOM—End of Month

EOY—End of Year

ES—Equipment Specialist

ETIMS—Enhanced Technical Management System

FCRN—Funds Classification Reference Number

FFF—Form, Fit, Function

FM—Financial Management

FY—Fiscal Year

G004L—Job Order Master Production System

G&A—General and Administrative

GFE—Government Furnished Equipment

GSA—General Services Administration

GSD—General Support Division

HQ—Headquarters

HQ AFMC—Headquarters Air Force Materiel Command

IAW—In Accordance With

ID—Identification

IPT—Integrated Product Team

IR—Incident Report

ITA—Interface Test Adapter

J025A—Automated Production Order System

J&A—Justification & Approval

JDRS—Joint Deficiency Reporting System

JON—Job Order Number

LRU—Line Replaceable Unit

$M—Millions

MAJCOM—Major Command

MFT—Metrics Financial Tool

MIP—Material Improvement Plan

MIPR—Military Interdepartmental Purchase Request

MP&E—Maintenance Planning & Execution

MPC—Maintenance Priority Code

MPST—Maintenance Engineering Project Sustainment Tool

MSD—Material Support Division

MXW—Maintenance Wing

NIIN—National Item Identification Number

NLT—No Later Than

NSN—National Stock Number

OC—Oklahoma City (Tinker AFB)

OC-ALC—Oklahoma City – Air Logistics Complex

OCP—Organizational Change Proposal

OCR—Operational Change Request

ODL—Open Document Listing

OFP—Operational Flight Programs

OL—Operating Location

OLRV—On Line Report Viewer

OMEI—Other Major End Item

OoC—Out of Cycle

OO—Ogden (Hill AFB)

OPR—Office Primary Responsibility

PBR—Program Budget Review

PCN—Program Control Number

PEC—Program Element Code

PK—Contracting

PMD—Program Management Directive

PM—Program Managers

POC—Point of Contact

PON—Purchase Order Number

POM—Program Objective Memorandum

PQDR—Product Quality Deficiency Report

PR—Purchase Request

PWS—Performance Work Statement

R2D2—Requirements Review and Deport Determination

RA—Resource Advisor

RAD—Requirements Approval Document

RCC—Resource Control Center

RDS—Records Disposition Schedule

RDT&E—Research, Development, Test & Evaluation

RFQ—Request for Quote

RG—Requirement Generator

RGC—Repair Group Category

RGC-S—Repair Group Category - Software

ROM—Rough Order Magnitude

RPM—Repair Program Manager

RTOK—Re-Test OK

SASPO—Strategic Alternate Sourcing Program Office

SCC—Software Control Center

SCM—Supply Chain Management

SCMG—Supply Chain Management Group

SCMS—Supply Chain Management Squadron

SCMW—Supply Chain Management Wing

SOR—Source of Repair

SOS—Source of Supply

SOW—Statement of Work

SPM—System Program Manager

SPO—System Program Office

SPON—Sponsor Order Number

SRU—Shop Replaceable Unit

STR—Software Trouble Report

SWDR—Software Deficiency Report

SWEG—Software Engineering Group

SWO—Software Organization

TAR—Tri-Annual Review

TCTO—Time Compliance Technical Order

TDY—Temporary Duty

T.O.—Technical Order

TPS—Test Program Set

ULO—Unliquidated Obligations

UUT—Units Under Test

WA—Work Authority

WR—Warner Robins (Robins AFB)

Terms

Accountable Forms—Forms that the Air Force stringently controls and which cannot be released to unauthorized personnel, since their misuse could jeopardize DOD security or result in fraudulent financial gain or claims against the government.

Administrative Change—Change that does not affect the subject matter content, authority, purpose, application, and/or implementation of the publication (e.g., changing the POC name, office symbol(s), fixing misspellings, etc.)

Approval Authority—Senior leader responsible for contributing to and implementing policies and guidance/procedures pertaining to his/her functional area(s) (e.g., heads of functional two-letter offices).

Authentication—Required element to verify approval of the publication; the approval official applies his/her signature block to authenticate the publication. The signature block includes the official's name, rank, and title (not signature).

ATE Hardware—An assembly of rack and stack test instruments, which are interconnected and computer controlled by the TPS, used to test the UUT. Software changes can be made to the ATE. ATE can also be used at the organizational level to test systems on the aircraft.

CSS—Includes the operating system, the Test Executive, Self-Test software, Calibration Test software, software drivers, and ATE unique software tools.

TPS——The software, firmware/hardware such as Interface Test Adapters (ITAs), and instructions used to diagnose performance failures of the UUT, and subsequent to repair completion, certify the UUT is fit for use. Software changes are made to the software portion of the TPS intended to diagnose, repair, and/or certify CSAG-S UUT Budget 8 coded item.

UUT—The item requiring repair and certification as fit for use. Items can include; Avionics Shop Replaceable Unit (SRU) or Line Replaceable Unit (LRU). CSAG-S UUT funded software is the software that operates within the SRU/LRU to control a function of that UUT (Core Content, specific guidance and procedures) (Not used in Air Force Policy Directive (AFPD))

Attachment 2

SAMPLE AUDIT CHECKLIST

Figure A2.1. Sample Audit Checklist.

	TPS Program Project Audit Checklist POC: Strategic Sourcing Program Office (Bolded items = Majors)	Category	Yes	No	NA	Comments
1.0	Project Validation Review					
1.1	Does this Project meet the Scope and Requirements of the TPS project funding program and been documented in MPST	Major				
1.2	Has the NSN of the Project been Identified in MPST and budget code verified?	Major				
1.3	Does the project accurately portray the correct project category i.e.(TPS)	Minor				
1.4	Has the project documentation been properly prepared and stored in the document repository of MPST	Minor				
1.5	Does the project have a clearly stated purpose in MPST	Minor				
1.6	Does the project contain a clearly written description in MPST	Minor				
1.7	Ensure that all Engineer, Program Manager, Item Manager, Equipment Specialist, and Contracting POC information is documented in MPST	Minor				
1.8	Does the project have an established TPS PCN and ensured requirements were file maintained within MP&E, and meet R2D2.	Major				
1.9	Has a Review Panel been established at the Supply Chain Management Groups (SCMGs) DoEs to evaluate, validate, and approve projects at site level	Major				
1.10	Has the project been coordinated by all approving officials(TPS: SCMS Engineering Flight Chief, SCMG Director of Engineering, SCM CC/DD, Resource Advisor, and 448 SCMW FM).	Minor				
1.11	Has an ALC review been conducted to ensure continuity and completion of project document and submittals	Minor				
1.12	Has documentation been identified to the TPS dificiency at the NSN level typically a DR, IR, or SWDR by weapon system	Minor				
1.13	Has the ALC completed Technical and Functional review and completed checkpoints in MPST	Major				
1.14	Does a Program Office review exist for verifying and validating project scope, documentation, coordination and ALC review	Minor				
1.15	Has the Program office completed Technical and Functional review and completed checkpoints in MPST	Major				
1.16	If a project was generated after the FY Data Call Final Submission Deadline, is it annotated as an Out-of-Cycle	Minor				
1.17	Ensure radio button for DoE report is selected Yes after funded, this is to make sure the project appears on the monthly DoE report for status	Major				
2.0	Acquisition Planning Meeting					
2.1	Has Project Acquisition Planning Meeting occurred and a timeline has been established, provide meeting minutes	Major				
4.0	Project File Maintenance					
4.1	Has the project been file maintained after validation and funding	Major				
4.2	Has a PR# and Date been recorded into MPST	Minor				
4.3	Has a Contract# as well as a Beginning & End date been established in MPST	Minor				
4.4	Has monthly execution status to include hours applied, DRs opened/closed, dollars expensed throughout project completion been provided					
4.4	Has a report for estimated dollars and hours needed to complete work on open JONs (AFMC Form 206s, Production #, T-jobs, etc.) prior to end of year closeout been provided	Minor				
4.5	Ensure the hours file maintained in MP&E were used when projecting hours in MPST	Major				